17th IMSO Answers

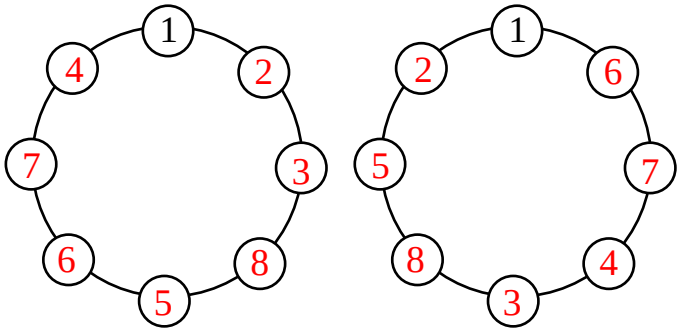
SHORT ANSWER

1.	25	**2.**	75°	**3.**	6	**4.**	4	**5.**	860
6.	25	**7.**	998	**8.**	14	**9.**	1199976	**10.**	24
11.	252	**12.**	22	**13.**	3	**14.**	21	**15.**	31
16.	180	**17.**	225	**18.**	35	**19.**	960	**20.**	T2 and T3
21.	4	**22.**	11	**23.**	519	**24.**	3750	**25.**	704

ESSAY PROBLEMS

1. 55

2.



3.

1st	2nd	3rd	4th
D	C	A	B

4. 4390

5. 32

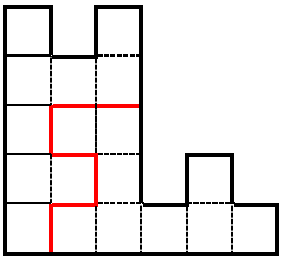
6. 30

7. 158

8. 31

9. 21

10.



, 40

11. Yes, the final points are 3, 4, 5, 6 and 7 points and one example is as following:

	I	II	III	IV	V	points
I		1	3	0	3	7
II	1		1	3	1	6
III	0	1		3	1	5
IV	3	0	0		1	4
V	0	1	1	1		3

12. 23, the diagram below shows one way to fill the grid.

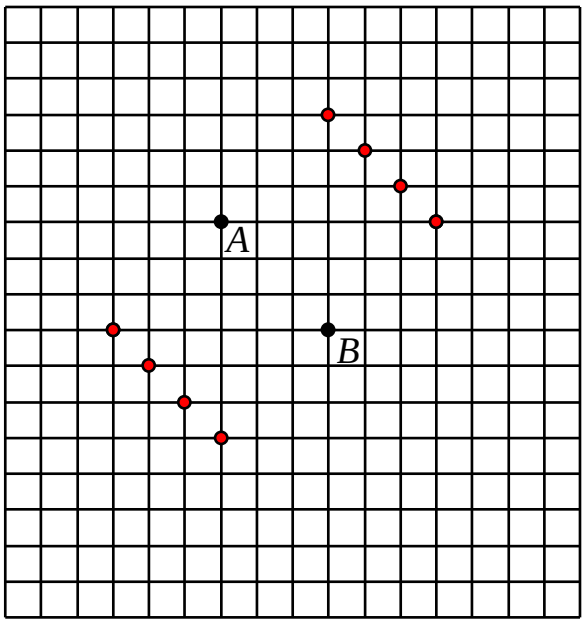
0	0	0	0	0	**0**
0	0	0	0	1	**1**
0	0	0	0	5	**5**
0	0	0	6	1	**7**
2	3	4	0	1	**10**
2	**3**	**4**	**6**	**8**	

13. 174

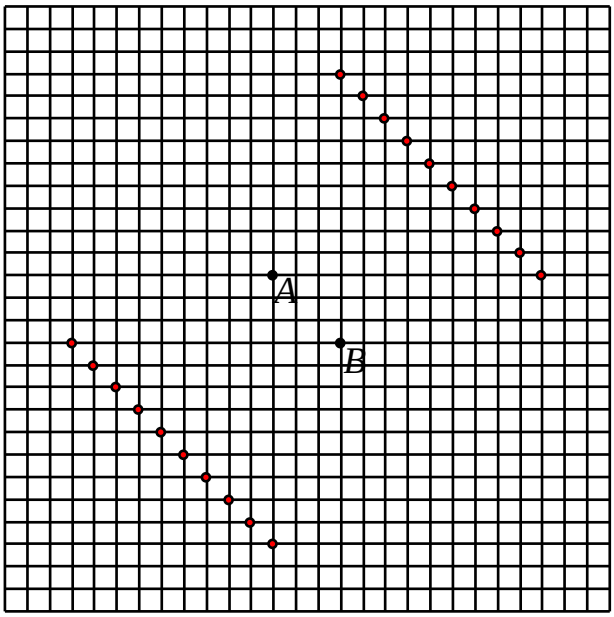
EXPLORATION PROBLEMS

1.

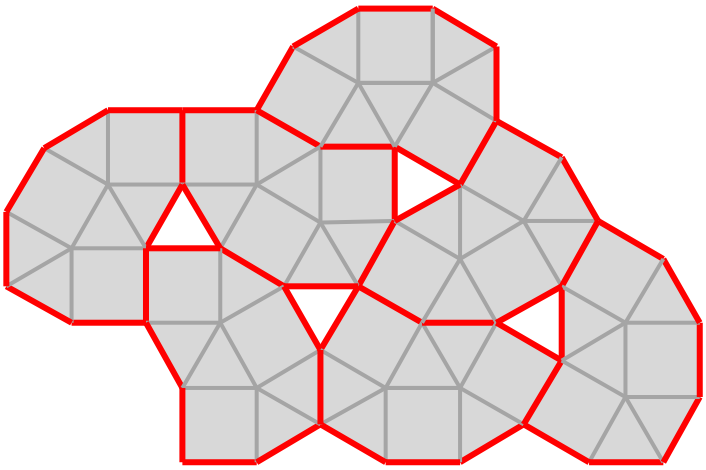
(a) There are 8 points as shown in the diagram below:



(b) There are 20 points as shown in the diagram below:

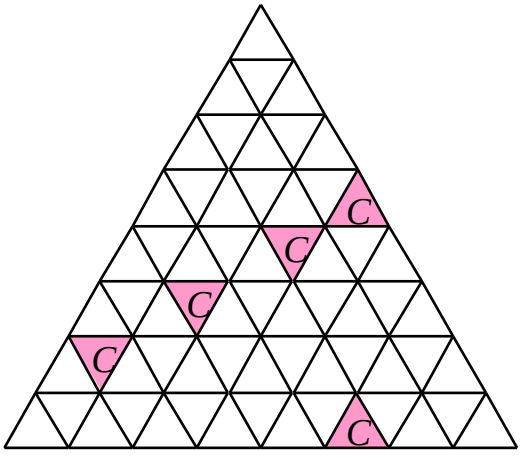


2.

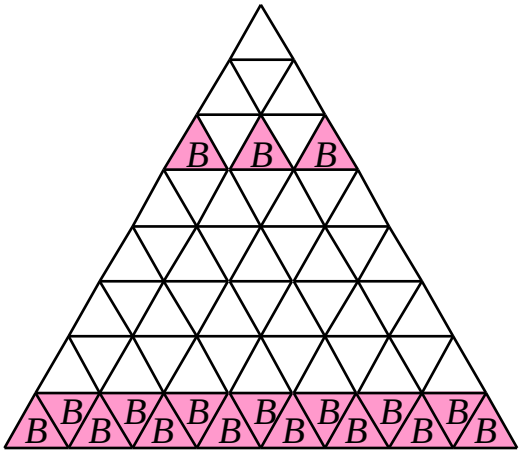


3.

(a) 5 castles are placed as shown in the diagram below:

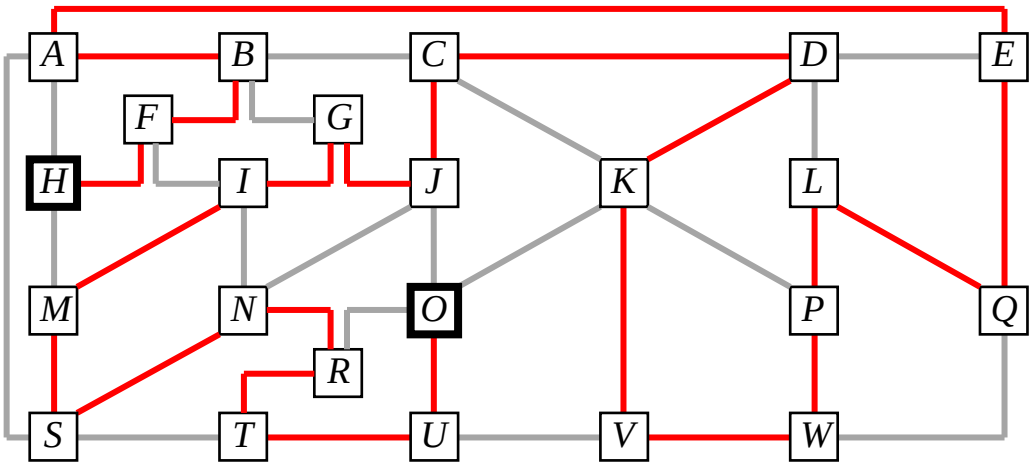


(b) 18 bishops are placed as shown in the diagram below:



4.

She can wait for him along the road between towns C and D.
One possible route as shown in the diagram below:



5.

(a) The minimum value of k for which Anna has a winning strategy is 6.

(b) If Anna has a winning strategy for $k=6$, then Boris can choose $k-1$ numbers such as 1, 2, 3, 5 and 7, so that Anna cannot guarantee a win.

6.

There are 5 possible products:

$$2+100+\underbrace{1+1+\ldots+1}_{98\text{ terms}}=200=2\times100\times\underbrace{1\times1\times\ldots\times1}_{98\text{ terms}}$$

$$4+34+\underbrace{1+1+\ldots+1}_{98\text{ terms}}=136=4\times34\times\underbrace{1\times1\times\ldots\times1}_{98\text{ terms}}$$

$$10+12+\underbrace{1+1+\ldots+1}_{98\text{ terms}}=120=10\times12\times\underbrace{1\times1\times\ldots\times1}_{98\text{ terms}}$$

$$4+4+7+\underbrace{1+1+\ldots+1}_{97\text{ terms}}=112=4\times4\times7\times\underbrace{1\times1\times\ldots\times1}_{97\text{ terms}}$$

$$2+2+3+3+3+\underbrace{1+1+\ldots+1}_{95\text{ terms}}=108=2\times2\times3\times3\times3\times\underbrace{1\times1\times\ldots\times1}_{95\text{ terms}}$$